

Test 24: Dirac string flux verification

Verdict: PASS | Generated: 2026-09-15 01:26:46 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

EXACT: plaquette flux = $2\pi q \pmod{2\pi}$; tolerance = round-off accumulation across the lattice.

METHOD: EXACT: plaquette flux = $2\pi q \pmod{2\pi}$; tolerance = round-off accumulation across the lattice.

Test 24 HARDCORE: 4-config full scan on 96x96, worst empty flux 8.88e-16, hermiticity true.

Result [HARDCORE pass 2]: 5 sub-checks, 0 failed ? PASS

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Plots

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 24 --no-two-pass`

Independent verification guide: `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

[01:26:05.603] [+5956.631s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=4.00, open, 1 vortices, model=:dirac

[01:26:18.836] [+5969.864s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=4.00, open, 1 vortices, model=:dirac

FULL COMPUTATION LOG

[01:26:28.248] [+5979.276s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=4.00, open, 1 vortices, model=:dirac

[01:26:37.309] [+5988.338s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=4.00, open, 1 vortices, model=:dirac

??

TEST 24 (pass 2, HARDCORE): Dirac-string flux ? exhaustive scan

CONSOLE CAPTURE

Full plaquette scan x4 configs: 2 positions x{+1, ?1, +0.3}

??

Lattice: 96x96 (N=9216 sites) ? full plaquette scan x4 configs

cfg A (ix=32, iy=48, q=+1.0): vortex 1/1, empty 9024/9024 ? OK

cfg B (ix=64, iy=24, q=+1.0): vortex 1/1, empty 9024/9024 ? OK

cfg A? (ix=32, iy=48, q=-1.0): vortex 1/1, empty 9024/9024 ? OK

cfg A frac (ix=32, iy=48, q=+0.3): vortex 1/1, empty 9024/9024 ? OK

Max |flux| over ALL empty plaquettes and configs: 8.88e-16

??

scan A q=+1.0 PASS ? 9025/9025 plaquettes OK

scan B q=+1.0 PASS ? 9025/9025 plaquettes OK

scan A? q=-1.0 PASS ? 9025/9025 plaquettes OK

scan A frac q=+0.3 PASS ? 9025/9025 plaquettes OK

hermiticity (all) PASS ? verify_hermitian per config

Test 24b [HARDCORE pass 2]: 5 sub-checks, 0 failed ? PASS